HULOG: COSTING, PRICING & MARKET SIZING (TAM SAM SOM)

PROJECT DOSSIER
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The Idea: Autonomous IoT gate drop vault with a heavy-duty 12V solenoid lock. Couriers deposit parcels using a **single-use, expiring 6-digit PIN / dynamic QR** without needing an app. Optical IR sensor & door limit switch verify package entry and lock closure to stop parcel theft.

1. ITEMIZED COST BREAKDOWN

COGS: ₱2,000.00

A. MATERIALS SUBSYSTEM	PRICE
ESP32 DevKit V1 (Wi-Fi/BLE Controller)	₱240.00
12V 0.8A Electromagnetic Solenoid Lock	₱190.00
12V 2A DC Power Adapter Supply	₱115.00
0.96" Monochrome I2C OLED Display	₱110.00
4x4 Matrix Membrane Keypad	₱65.00
LM2596 DC-DC Buck Converter (12V to 5V)	₱55.00
MC-38 Magnetic Door Reed Switch	₱55.00
1-Channel 5V Optocoupler Relay	₱45.00
Optical IR Parcel Presence Sensor	₱25.00
5V Active Piezo Audio Buzzer	₱20.00
Internal Wire Harness, DC Jacks & Connectors	₱80.00
Subtotal — Electronics Subsystem:	**₱1,000.00**
1.2mm Galvanized Metal Enclosure (Laser-cut)	₱500.00
Outdoor Anti-Rust Powder Coating	₱120.00
Continuous Piano Hinges, Gaskets & Screws	₱80.00
Subtotal — Direct Materials (BOM):	**₱1,700.00**

B. MANPOWER / LABOR (2.0 HRS @ ₱90/HR)	COST
Circuit Wiring, Buck Tuning & Soldering (45m)	₱67.50
Solenoid Lock & Mechanical Integration (45m)	₱67.50
Firmware Flashing & QA Door Cycle Test (30m)	₱45.00
Subtotal — Direct Manpower:	**₱180.00**

C. OVERHEAD & PACKAGING	COST
Packaging Box, Foam Cushioning & Manual	₱65.00
Consumables (Solder, heat shrink, zip ties)	₱25.00
Shop Electricity & Tool Wear Allocation	₱30.00
Subtotal — Overhead & Packaging:	**₱120.00**

2. PRICING & PROFITABILITY

PRICE: ₱3,499.00

FINANCIAL METRIC	AMOUNT (PHP)	MARGIN
Retail Price (Base Unit)	**₱3,499.00**	—
Unit COGS (Materials+Labor+Overhead)	₱2,000.00	—
Gross Profit per Unit	**₱1,499.00**	**42.8%**
Allocated OPEX (Marketing, shipping, warranty)	₱500.00	—
Net Profit per Unit	**₱999.00**	**28.5%**
Pro Tier (w/ Li-Ion Battery Backup)	**₱4,499.00**	45.5%
Optional SaaS (Dorm / Boarding House)	**₱199.00/mo**	>85%

3. MARKET SIZING (TAM • SAM • SOM)

TAM: Total Addressable Market — ₱38.5 Billion

Scope: Total Philippine online shopping households.
• ~11.0M active e-commerce buying households nationwide.
• **Formula:** $11,000,000\text{ households} \times \text{₱3,500} = \mathbf{₱38.5\text{B}}$ (~$687M USD).

SAM: Serviceable Addressable Market — ₱5.25 Billion

Scope: Mega Manila & Key Metros (NCR, Cavite, Laguna, Rizal).
• Standalone gated homes, townhouses & student dorms with regular unattended delivery issues (~1.5M target dwellings).
• **Formula:** $1,500,000\text{ units} \times \text{₱3,500} = \mathbf{₱5.25\text{B}}$ (~$93.7M USD).

SOM: Serviceable Obtainable Market — ₱35.0 Million

Scope: 3-Year Realistic Sales Goal (~0.65% share of SAM).
• **10,000 units** sold via U-Belt dorms & HOA partnerships.
• **Formula:** $10,000\text{ units} \times \text{₱3,500} = \mathbf{₱35.0\text{M}}$ hardware.
• *Plus SaaS:* 1,200 dorms @ ₱199/mo = **+₱2.86M/yr**.

COGS: ₱2,000.00 | **RETAIL:** ₱3,499.00 | **GROSS MARGIN:** 42.8%

SOM: ₱35,000,000.00 (10k Units)